Identifying Fraudulent and Non-Compliant Airbnb Listings

Across New York City Boroughs

ISM5560

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1. Problem Definition

The purpose of this project is to determine whether fraudulent or non-compliant Airbnb listings are more concentrated in certain New York City boroughs. Fraudulent or unsafe listings may occur when short-term rentals are located in buildings with active Housing Maintenance Code violations, potentially presenting unsafe living conditions, overcrowding, or illegal conversions.

This project examines the question:
Are some NYC boroughs more likely to contain fraudulent or non-compliant Airbnb listings?

This information provides actionable insights to stakeholders through:
- Regulators, who must target enforcement
- Airbnb, which must maintain platform trust
- Guests, who rely on safe accommodations
- Legitimate hosts, who depend on a fair marketplace

By linking Airbnb coordinates to NYC Housing Code violations, we identify boroughs where the risk of non-compliance is highest.

2. Data Understanding & Preparation

2.1 Airbnb Listing Data

Contains ~ 49,000 listings with:
- latitude, longitude
- price, room_type, minimum_nights
- borough information
- host information & review metrics

2.2 HPD Violations Data

Contains ~10.5 million building-level violations with:
- latitude, longitude
- violation class & description
- Notice of Violation Type date and description

2.3 Data Cleaning and Preparation Steps

1. Derive crucial information about each dataset, such as datatypes, null counts, columns, and duplicates
    - Utilized .info(), .duplicated().sum(), and ensured date values for reviews and Notice of Violation dates were converted to Pandas datetime format.
    - Found the most recent and oldest review dates to filter HPD Violations Data.

- Dropped unnecessary columns from HPD Violations Data.

2. Coordinate Standardization
   - Coordinates varied from 3–6 decimals.
   - We rounded all latitude/longitude values to 4 decimals (~11 meters) to match buildings consistently, then removed duplicate longitude/latitude columns.
   - Removed any rows with missing longitude and latitude values, as these were meant to be used to merge the datasets.
3. Merge datasets into a unified dataframe for analysis
   - Merged datasets on standardized coordinates, and conducted further cleaning by removing more columns that were not relevant for analysis.
   - Identified null/NaN values, and replaced these with 'None', as they were listings that did not have a match from the HPD Violations dataset (i.e. non-fraud listings).
4. Spatial Matching: Fraud Flag Construction
   - A listing was marked as fraudulent/non-compliant if its rounded coordinates matched those of a building with code violations.

     fraud_flag = 1 (matched)

     fraud_flag = 0 (not matched)
5. Borough Aggregation

   For each borough, we calculated:

   - total listings
   - total fraudulent listings
   - fraud rate (%)

This created the final dataset used in the three analyses.

```
merged.head()
```

	neighbourhood_group	latitude	longitude	room_type	price	minimum_nights	number_of_reviews	reviews_per_month	calculated_host_listings_count	availability_3
0	Manhattan	40.766610	-73.988100	Entire home/apt	144.0	30	9	0.24	139	3
1	Manhattan	40.750764	-73.994605	Entire home/apt	187.0	2	6	1.67	1	3
2	Manhattan	40.835600	-73.942500	Private room	120.0	30	156	1.38	2	3
3	Manhattan	40.751120	-73.978600	Entire home/apt	85.0	30	11	0.24	133	3
4	Manhattan	40.759950	-73.992960	Entire home/apt	115.0	30	5	0.16	139	2

```
35]: merged.head()
```

35]:

calculated_host_listings_count	availability_365	number_of_reviews_ltm	beds	Class	NOVIssuedDate	CurrentStatus	NovType	ViolationStatus	RentImpairing	fraud_flag
139	364	2	1	None	None	None	None	None	None	0
1	343	6	2	None	None	None	None	None	None	0
2	363	12	1	None	None	None	None	None	None	0
133	335	3	1	None	None	None	None	None	None	0
139	276	2	1	None	None	None	None	None	None	0

3. Analysis

Chi-Square Analysis

- Dependent variable:
  - fraud_flag (0 = compliant, 1 = fraudulent)
- Independent variable:
  - borough (categorical with 5 levels)

This type of analysis was chosen to measure how the observed frequencies differ from expected frequencies if the variables were independent. We used this to test whether borough and fraud status are independent of each other. Fraud is a categorical outcome, and borough is a categorical variable. Level: Borough level, because we're comparing counts of fraud vs. non-fraud listings across borough categories.

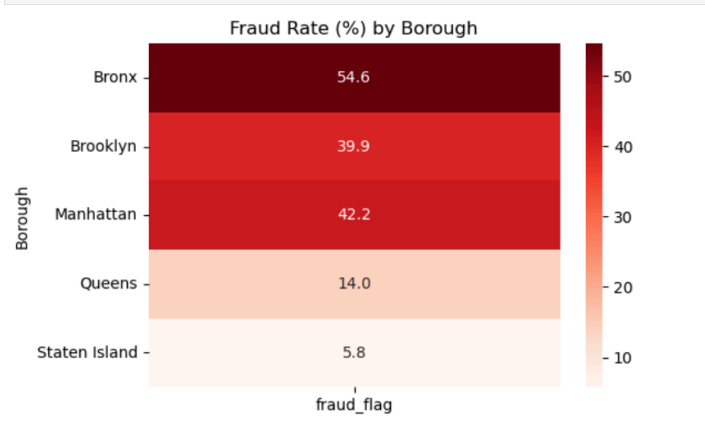
H0: Fraud occurrence is independent of Borough
Ha: Fraud occurrence is not independent of Borough

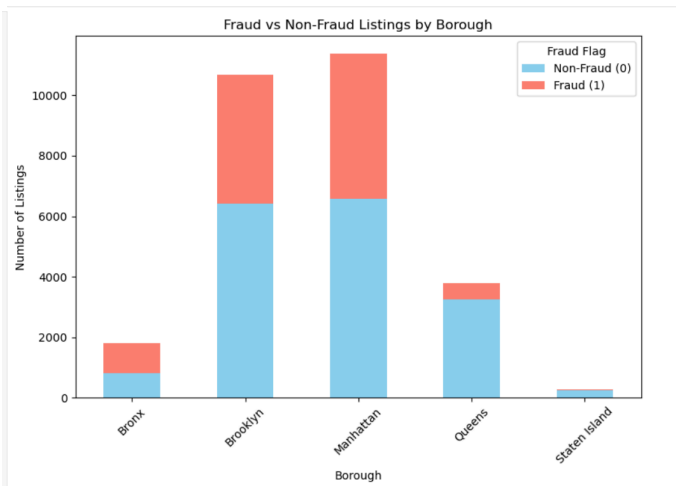
```
import scipy.stats as stats

contingency = pd.crosstab(merged['neighbourhood_group'], merged['fraud_flag'])

chi2, p, dof, expected = stats.chi2_contingency(contingency)

print("Chi-square statistic:", chi2)
print("Degrees of freedom:", dof)
print("p-value:", p)
```

```
Chi-square statistic: 1358.3730788081339
Degrees of freedom: 4
p-value: 7.339420777695151e-293
```

```
print(contingency)
contingency_norm = contingency.div(contingency.sum(axis=1), axis=0)
print(contingency_norm)
```

```
fraud_flag              0     1
neighbourhood_group
Bronx                 820   986
Brooklyn             6416  4265
Manhattan            6584  4804
Queens               3252   530
Staten Island         260    16
fraud_flag                 0         1
neighbourhood_group
Bronx                0.454042  0.545958
Brooklyn             0.600693  0.399307
Manhattan            0.578152  0.421848
Queens               0.859863  0.140137
Staten Island        0.942029  0.057971
```





Results: Chi-square statistic = 1358.37, df = 4, p-value ≈ 0. Fraud occurrences are highly disproportionate across boroughs (relative to borough size). Our Chi-Square analysis proved to be significant, as our p-value is less than the 0.05 threshold. We have enough evidence to reject the null hypothesis that fraud occurrence is independent of borough. Bronx has the highest percentage (54.6%), and Staten Island the lowest (5.8%). In the 'Fraud vs. Non-Fraud Listings by Borough' stacked bar chart, we can see that Bronx also has the lowest number of listings overall. This suggests Bronx listings are disproportionately risky.

Levene's Test (Variance Equality)

- Dependent variable:
  - price (continuous Airbnb listing price)
- Independent variable:
  - borough (categorical with 5 levels)

Levene's was used to test for equal variance across all boroughs' Airbnb listing prices, due to the size of each borough being different. The result of this test would determine which type of ANOVA test should be used in the next step, as typical One-Way ANOVA assumed equal variance.

H0: All boroughs have equal variance in price.
Ha: At least one borough has a different variance in price.

```python
# Levene's to test for variance across groups
from scipy.stats import levene

groups = [merged.loc[merged['neighbourhood_group']==g, 'price']
          for g in merged['neighbourhood_group'].unique()]
stat, p_val = levene(*groups, center='median')

print(f"Levene's test statistic: {stat:.3f}, p-value: {p_val:.3f}")
```

```
Levene's test statistic: 3.734, p-value: 0.005
```

Results: p-value 0.005<0.05, therefore, we have enough evidence to reject the null that all boroughs have equal variance in price. Our result deterred us from using One-Way ANOVA, and we ultimately chose Welch's.

Welch's ANOVA

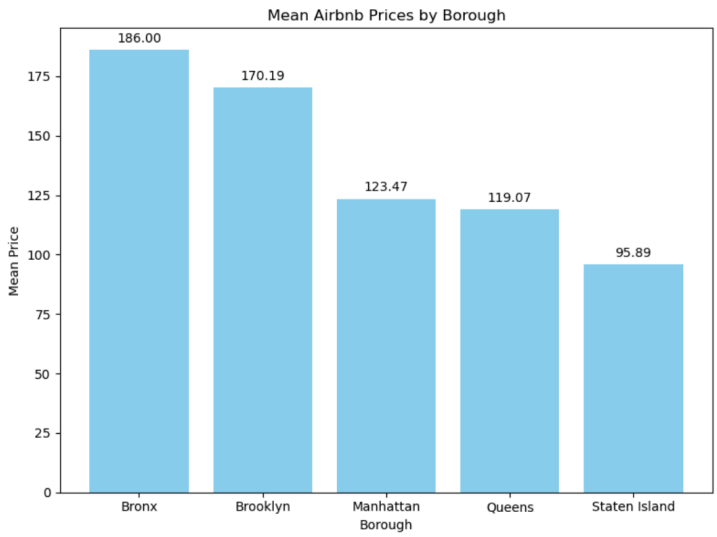
- Dependent variable:
  - price (continuous Airbnb listing price)
- Independent variable:
  - borough (categorical with 5 levels)

Price is continuous, borough is categorical. ANOVA is the right test to compare means across multiple groups. Welch's version was chosen because equal variances were violated. Welch's ANOVA was conducted to test whether boroughs differ significantly in average listing price. Level: Borough level, because we're comparing borough-wide averages of price, not individual listing features.

H0: Borough's do not differ in mean Airbnb listing price
Ha: At least one borough differs in mean Airbnb listing price

```python
# Welch's ANOVA

import statsmodels.stats.oneway as oneway

groups = [merged.loc[merged['neighbourhood_group']==g, 'price']
          for g in merged['neighbourhood_group'].unique()]
welch = oneway.anova_oneway(groups, use_var='unequal')
print(welch)
```

```
statistic = 184.9313069386119
pvalue = 6.074510734587779e-135
df = (4.0, np.float64(1997.147942285058))
df_num = 4.0
df_denom = 1997.147942285058
nobs_t = 27933.0
n_groups = 5
means = [186.00289779 170.18621852 123.46589106 119.06884058  95.88648948]
nobs = [11388. 10681.  3782.   276.  1806.]
vars_ = [  39197.37129221 1893786.88151792   11069.01007669    8519.15524374
       57443.87353482]
use_var = unequal
welch_correction = True
tuple = (np.float64(184.9313069386119), np.float64(6.074510734587779e-135))
```



Result: Significant differences are found in mean Airbnb listing prices across boroughs. Since our p-value was less than 0.05 and Welch's F statistic was extremely high (184.9), we have enough evidence to reject the null that boroughs do not differ in mean listing price. This test was conducted with 27933 observations across 5 groups (boroughs). Bronx has the highest mean price of 186, suggesting that boroughs with higher fraudulent listings may have higher prices.

Logistic Regression

The purpose of conducting this analysis was to determine which listing features predict fraud risk. Fraud is binary (fraud vs. non-fraud), and logistic regression allows us to quantify the effect of multiple predictors simultaneously. Level: Listing level, because we wanted to move beyond borough averages and measure how individual features influence fraud odds. SMOTE was used to resample and improve model performance to meet the goals of this project (first table in the figure below is before using SMOTE, the second table below Odds Ratio table is after using SMOTE).

H0: Predictor variables (room type, beds, borough, reviews, price) have no effect on fraud odds (odds ratio = 1).
Ha: At least one predictor significantly affects fraud odds (odds ratio does not equal 1).

```
[[4481  719]
 [2144 1036]]
              precision    recall  f1-score   support

           0       0.68      0.86      0.76      5200
           1       0.59      0.33      0.42      3180

    accuracy                           0.66      8380
   macro avg       0.63      0.59      0.59      8380
weighted avg       0.64      0.66      0.63      8380

                            Feature  Coefficient  Odds_Ratio
12            room_type_Private room     0.279116    1.321961
6                               beds     0.192589    1.212384
2                  number_of_reviews    -0.000101    0.999899
1                     minimum_nights    -0.000494    0.999506
4                   availability_365    -0.000559    0.999441
0                              price    -0.002715    0.997288
5      calculated_host_listings_count    -0.006301    0.993719
8       neighbourhood_group_Manhattan    -0.124746    0.882721
3                  reviews_per_month    -0.144539    0.865421
7        neighbourhood_group_Brooklyn    -0.292909    0.746090
11               room_type_Hotel room    -0.676404    0.508442
13              room_type_Shared room    -1.064269    0.344980
9          neighbourhood_group_Queens    -1.850430    0.157170
10  neighbourhood_group_Staten Island    -2.287302    0.101540
[[2555 2645]
 [ 923 2257]]
              precision    recall  f1-score   support

           0       0.73      0.49      0.59      5200
           1       0.46      0.71      0.56      3180

    accuracy                           0.57      8380
   macro avg       0.60      0.60      0.57      8380
weighted avg       0.63      0.57      0.58      8380
```

Results: Logistic regression coefficients showed significant odds ratios for multiple predictors. Fraud risk is driven by identifiable listing features. The key predictors that increase likelihood for fraudulent listings are private rooms, more beds, and being located in the Bronx (consistent with our previous Chi-Square result). The odds of fraud in listings that are private rooms are significantly higher compared to entire homes or hotel/shared rooms. Each additional bed increases the likelihood of fraud, suggesting overcrowding or illegal conversions. The key predictors that decrease the likelihood for fraudulent listings are more reviews, higher prices, hotel/shared rooms, and Queens/Staten Island listings.

*Although Bronx has the highest percentage of fraud and highest average pricing (according to Welch's ANOVA result above), when aggregated at borough-level, Bronx's average listing price is relevant to group averages and proportions, whereas this model describes individual listings.

Model 1 (Accuracy-focused; pre-SMOTE)
- Accuracy = 66%
- Recall = 33% (weak at catching fraud cases)

- Precision = higher, but misses most fraud.
- Good at predicting non-fraud, but poor at identifying fraud.

Model 2 (Recall-focused; post-SMOTE)
- Accuracy = 57%
- Recall = 71% (stronger at catching fraud cases)
- Precision = lower, but far fewer fraudulent listings slip through.
- Better for enforcement/regulation, since missing fraud is more costly than false alarms.

Ultimately, we chose to use Model 2, post-SMOTE as our model for prediction and interpretation. Our specific business goals aligned with prioritizing recall (catching fraud cases), although there was an accuracy tradeoff.

- Dependent variable:
  - fraud_flag (0 = compliant, 1 = fraudulent)
- Independent variable:
  - borough (categorical with 5 levels: Bronx, Brooklyn, Manhattan, Queens, Staten Island; Bronx used as baseline)
  - neighbourhood_group_Staten Island
  - neighbourhood_group_Queens
  - room_type_Shared room
  - room_type_Hotel room
  - neighbourhood_group_Brooklyn
  - reviews_per_month
  - neighbourhood_group_Manhattan
  - calculated_host_listings_count
  - price
  - availability_365
  - minimum_nights
  - number_of_reviews
  - beds
  - room_type_Private room

4. Interpretation & Insight

4.1 Key Findings

Our results consistently show that fraudulent or non-compliant Airbnb listings are not evenly distributed across NYC boroughs.

Key findings include:

- The Bronx had the highest fraud rate relative to total listings, suggesting dense clusters of code violations affecting short-term rentals.
- Manhattan also showed an elevated fraud concentration, likely driven by older building stock and stricter housing regulations that many listings may violate.
- Staten Island had the lowest fraud rate, indicating far fewer listings linked to HPD violations.

- Brooklyn and Queens fell in the middle, showing moderate but significant levels of potential non-compliance.
- All three statistical tests (Chi-Square, Welch's ANOVA, and Logistic Regression) confirmed that borough significantly predicts fraud likelihood.

These results reflect meaningful geographic disparities in building condition, enforcement levels, and housing density across boroughs.

4.2 Managerial & Policy Implications

- Airbnb should increase monitoring of high-risk boroughs.
- Regulators can target inspections where violations overlap with short-term rentals.
- Hosts may face higher scrutiny in boroughs with repeat violations
- Guests benefit from clearer information on neighborhood safety.

4.3 Limitations

- *Spatial Matching Accuracy*: The fraud flag depends on rounding coordinates to four decimals (because this level of precision ,~11 meters, balances computational feasibility with the need for block-level spatial matching across large datasets) which can create false positives and false negatives when matching Airbnb listings to HPD buildings. This reduces confidence in the fraud variable. Future work should use parcel IDs or building footprints for more accurate matching.
- *Building-Level vs. Unit-Level Violations*: HPD violations apply to the entire building, not the specific Airbnb unit. A listing may be flagged even if its individual unit is compliant. Future research should incorporate DOB complaints, 311 data, or unit-level enforcement records when available.
- *Correlation Does Not Equal Causation*: Differences in average price or listing characteristics across boroughs should not be interpreted as causing fraud. Some patterns (e.g., Bronx having higher mean price and higher fraud rate) may be spurious due to small sample sizes or borough-specific factors.
- *Time Misalignment*: The Airbnb dataset and HPD violation records do not perfectly align in time. A building may have resolved violations before the listing date, or new issues may have occurred after. Time-aligned data would improve accuracy.
- *Omitted Variables:* Important factors such as zoning rules, building ownership, or prior enforcement actions were unavailable. Adding these datasets could strengthen the results.

4.4 Future Work

To strengthen the results, future work should:

- Use exact parcel IDs (BBL numbers) instead of rounded coordinates for precise matching.
- Integrate 311 complaint data, especially illegal conversions and unsafe building complaints.
- Add temporal analysis, such as:
    - Whether fraud increases during peak tourism seasons

    - Whether violations precede or follow Airbnb activity
- Evaluate host-level fraud patterns, such as hosts with multiple flagged listings across boroughs.
- Use geospatial clustering models (heatmaps, DBSCAN) to identify micro-neighborhood hotspots, not just borough-level patterns.

5. Conclusion

Our findings show clear evidence that fraudulent or non-compliant Airbnb listings are concentrated in specific boroughs, especially the Bronx and Manhattan. These areas contain older housing stock and higher volumes of HPD code violations, which increase the likelihood that Airbnb units operate in unsafe or non-compliant conditions.

These results provide direct, actionable value to stakeholders:

- Regulators can prioritize inspections and enforcement in boroughs with elevated fraud rates.
- Airbnb can strengthen automated screening, verification, and host documentation requirements for high-risk zones.
- Legitimate hosts benefit from improved fairness, as enforcement reduces competition from illegal operators.
- Guests gain enhanced safety and transparency, especially when choosing listings in historically higher-risk boroughs.

While matching limitations may introduce noise, the consistency across statistical tests strongly supports the conclusion: borough is a meaningful predictor of fraud, and targeted interventions can significantly improve housing safety and compliance across NYC.

6. *Executive Summary*

This project examines whether fraudulent or non-compliant Airbnb listings are disproportionately located in specific NYC boroughs. A listing was considered “fraudulent” if its coordinates matched a building with active Housing Maintenance Code violations recorded by NYC’s HPD. After standardizing coordinates, merging datasets (~49k Airbnb listings + ~10.5M violation records), and constructing a fraud indicator, we conducted Chi-Square tests, Welch’s ANOVA, and Logistic Regression to assess borough-level differences.

Across all methods, results show that fraudulent listings are unevenly distributed across NYC, with the Bronx and Manhattan showing the highest concentrations and Staten Island the lowest. These patterns likely reflect differences in building condition, age, and regulatory pressure across boroughs.

These findings offer clear stakeholder value:

- Regulators can deploy inspections more efficiently by focusing on boroughs with elevated fraud rates.
- Airbnb can direct trust and safety enforcement resources to boroughs where non-compliance risk is higher.

- Hosts and guests benefit from safer and more transparent short-term rental environments.

Although the analysis is limited by coordinate rounding, lack of unit-level violation data, and incomplete temporal alignment, the project provides a solid, data-driven foundation for improving oversight and creating targeted compliance strategies.

Sources:

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Open Data - HPD

Housing Maintenance Code Violations | NYC Open Data

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